

P1h: pushing Theorems M, B, C and the P1g conditions towards
 $\ell \rightarrow 0$

the arc $[0.143, 0.857]$ for $Z_N \neq 0$, the arc $[\frac{1}{6}, \frac{5}{6}]$ for U, V ,
and the exact obstructions below

research programme P1, eighth atom, 2026-09-24. Reviewed by Reviewer AD (judged correct with minor gaps;
gaps addressed 2026-09-24).

Notation and numbering of P1d, P1e (Theorem M, Lemmas Zk, R, A, H, B, B', O), P1f (Theorems B, C, Lemmas LS, LSp, head) and P1g (Theorem main, Lemmas B $^\pm$, I, Propositions S, L, A $^\pm$).
 $x = a/N$, $\ell = \log |2(1 - \zeta)| = \log(4 \sin \pi x)$, $g(x) = e^{-\ell} = 1/(4 \sin \pi x) \approx \sup |u_0|$. Labels: PROVED;
computer-assisted (finitely many Arb ball inequalities, checked by the named script, which fails loudly);
VERIFIED (numerics, not a proof); HEURISTIC; OPEN.

Correction to the brief. $\ell > 0$ is $x \in (0.08043, 0.91957)$, not $(\frac{1}{6}, \frac{5}{6})$: at $x = \frac{1}{6}$, $\ell = \log 2 = 0.693$.
The arc $[\frac{1}{6}, \frac{5}{6}]$ is well inside the good arc, and even inside the region $\ell > 0.484$ where the P1e method
can work in principle (Section 5).

Outcome.

- (a) **Theorem M' (PROVED, computer-assisted).** For all $N \geq 1$ and $\gcd(a, N) = 1$ with
 $a/N \in [x_1, 1 - x_1]$, $x_1 = 0.143$ (so $\ell \geq 0.55224$): $|Z_N(a/N)| \geq 0.02 e^{-0.005N}$. So $\Sigma_N \neq 0$ and
 $S_0 \neq 0$ there. This is $[\frac{1}{7} + \delta, \frac{6}{7} - \delta]$ with $\delta = 1.43 \cdot 10^{-4}$, and it contains $[\frac{1}{6}, \frac{5}{6}]$.
- (b) **Theorem main' (P1g conditions; PROVED, computer-assisted).** On $[\frac{1}{6}, \frac{5}{6}] \setminus \{\frac{1}{2}\}$: $|Z_N^\pm| \geq$
 $0.02 e^{-0.005N}$, $|\omega| \leq 7.96$, and $\min_i |G_i| \geq 0.0499$.
- (c) **Theorems B', C' (PROVED; computer-assisted through (a), (b)).** Every conclusion of
P1f Theorem B holds for $a/N \in [x_1, 1 - x_1]$ with $N \geq 17$, and for $a/N \in [\frac{1}{6}, \frac{5}{6}]$ with $N \geq 16$.
The conclusions of P1f Theorem C hold on $[\frac{1}{6}, \frac{5}{6}]$, $N \geq 16$. Hence:
 - the poles of G_0^T accumulate at every point of $\arg q \in [2\pi x_1, 2\pi(1 - x_1)]$;
 - the poles of U and V accumulate at every point of $\arg x \in [\frac{\pi}{6}, \frac{5\pi}{6}] \cup [\frac{7\pi}{6}, \frac{11\pi}{6}]$ (standing:
thm:model, thm:RJ).
- (d) **The landscape is not the obstruction (PROVED for each listed ℓ_1 , computer-assisted).**
Lemma LS(ℓ_1): every estimate of P1f Lemmas LS, LSp and head holds for $\ell \geq \ell_1$ and $N \geq$
 $N_1(\ell_1)$, with $N_1 \ell_1 \approx 9\text{--}19$ (Table 2). So P1f Theorem B holds at *every* ζ with $\ell > 0$, $N \geq N_1(\ell)$
and $Z_N(a/N) \neq 0$. The bound $T \geq 0.1499$ was tight only because the constants had been frozen
at $\ell_0 = 0.855$, $N = 16$. The true requirement is $N\ell \gtrsim 10$.
- (e) **The obstruction is Theorem M's hazard step.** It has three parts (Section 5):
 - the golden threshold $\ell > 0.4840$ ($x > 0.13295$) of the crude Lemma B;

- the crossover height $n_0(y)$ at which Lemma B takes over from exact seeds, which blows up as $y \rightarrow 0.6163$;
- the low-seed/direct-covering handover. The contour height $\eta \leq \log(y/g)/2\pi$ shrinks the certifiable disc of $1/7$ to $5 \cdot 10^{-6}$, while the direct covering needs about $3 \cdot 10^{-5}$. This is why the arc stops at 0.143 and not at $\frac{1}{7}$.

Below $\ell \approx 0.484$ the crude triangle bound cannot close the induction for any choice of y , and a structured (moment) hypothesis is needed (OPEN).

1 Region-dependent hazard bases

P1e uses one hazard base $y = \frac{1}{2}$, with $\sup |u_0| e^{2\pi\eta_i} \leq y$ on each region. Since $g(\frac{1}{6}) = \frac{1}{2}$ exactly, $y = \frac{1}{2}$ cannot serve $x < \frac{1}{5}$ with any $\eta > 0$. We use a base per region:

$$R_1 = [0.143, \frac{1}{6}] : y_1 = 0.59, \quad R_0 = [\frac{1}{6}, \frac{1}{5}] : y_0 = 0.52, \quad [\frac{1}{5}, \frac{1}{2}] : y = \frac{1}{2} \text{ (P1e, unchanged)}.$$

The hazard radius of a reduced j/n ($n \geq 10$) is $\varepsilon_n = 2y^n/n^2$ ($n \leq 52$) or $y^n/(4\tau)$ ($n \geq 53$), $\tau = \frac{1}{50}$, with y the base of the region containing j/n .

Lemma 1 (H_y ; PROVED, computer-assisted). *P1e Lemma H(1)–(4) holds verbatim on each region with its own base, provided $\sup_{\text{region}} |u_0| \rho \leq y$ for the ρ used. In particular this holds on R_0, R_1 with the parameters of Table 1.*

Proof. The proof of P1e Lemma H uses $y = \frac{1}{2}$ only through five numerical facts:

- (H1) $n^2 \varepsilon_n \leq \frac{1}{2}$;
- (H2) $y^{\max(60, 2m')}/(4\tau) \leq 2y^{m'}/m'^2$ for $10 \leq m' \leq 52$;
- (H3) $\varepsilon_n < \log(1/y)/(2n)$ and $f(\varepsilon_n) = \varepsilon_n y^{-1/(2n\varepsilon_n)} \geq 1/(4\tau)$ (for $n \geq 53$ this is $y^{-n} \geq 25n^2$);
- (H4) $y^{n_2+1}/(4\tau) < 1/(2n^2)$ for $n_2 = \lfloor 1/(2n\varepsilon_n) \rfloor - 1$. This replaces the P1e phrase “ $n_2 \geq 2560$ ”, which was specific to $y = \frac{1}{2}$.
- (H5) $y^{60}/(4\tau) < 10^{-6}$ and $12.5y^m < 1/(9m)$ for $m \geq 60$.

`P1h.hazard.check.py` checks (H1)–(H5) in Arb for $y \in \{0.5, 0.52, 0.59\}$ and $10 \leq n \leq 10^4$. Beyond that they are monotone, since $1/y \geq (1 + \frac{1}{53})^2$.

No region crossing. Suppose x is in one region and a reduced i'/m' with $\|mx\| < my^m/(4\tau)$, $m \geq 60$, lies across a boundary $b = P/Q \in \{\frac{1}{6}, \frac{1}{5}\}$ ($Q \leq 9$). Then $|i'/m' - b| \geq 1/(Qm') \geq 1/(9m) > 12.5y^m$ by (H5), which is impossible. The same holds for hazards of height $10 \leq n \leq 59$: $|j/n - b| \geq 1/(9n) > \varepsilon_n$. At the arc end $x_1 = 0.143$ there is nothing to cross. It is not a rational of small height, every seed of $[\frac{1}{7}, x_1)$ belongs to R_1 and is included in all of the certificates below, and $1/7$ itself lies at distance $1.43 \cdot 10^{-4} \gg y_1^{60}$ from the arc. So every hazard or low rational that is relevant for x has the base of x 's region, and the P1e proof applies with y replaced by that base. The low-seed disc radii used ($\geq 2 \cdot 10^{-5}$) exceed $y^{60}/(4\tau)$ by (H5). $\square$

Table 1: Region data (all Arb-checked). g = upper bound for $\sup |u_0|$ on the region (with factor $1 + 10^{-6}$). η = contour height of the crude Lemma B, with $g e^{2\pi\eta} \leq y$ checked in Arb. n_0 = first height of the crude Lemma B.

region	y	g	η	n_0 (crude from)	exact seeds	$\theta = e^\kappa y / \sqrt{1-y}$
$R_0 = [\frac{1}{6}, \frac{1}{5}]$	0.52	0.5	0.0062	67 (used: 151)	227, $10 \leq n \leq 150$	0.7544
$R_1 = [0.143, \frac{1}{6}]$	0.59	0.5762	0.0037	301	654, $10 \leq n \leq 300$	0.9260

2 Theorem M'

Proposition 2 (certificates; PROVED, computer-assisted). 1. Base and level- n data (*P1h_seeds.py*).

Covered: all 228 reduced $a/N \in [\frac{1}{6}, \frac{1}{5}]$ with $3 \leq N \leq 150$, and all 655 reduced $a/N \in [\frac{1}{7}, \frac{1}{6}]$ with $3 \leq N \leq 300$. Every one has $Z_N, Z_N^\pm \neq 0$ and all $G_i \neq 0$. On R_0 : $|Z| \geq 0.8244$, $|Z^\pm| \geq 0.5261$, $\text{margin} \geq 0.2946$, $|\omega| \leq 1.039$. On $[\frac{1}{7}, \frac{1}{6}]$: $|Z| \geq 0.5452$, $|Z^\pm| \geq 0.4610$, $\text{margin} \geq 0.4855$.

2. Low seeds (*P1e_lowseed.py*, *P1h_lowseed.py*; P1d Theorem 9 with P1e Lemma O; side $d > 0$):

seed	region side	r	η	$B \leq$	$\kappa_{\text{seed}} \leq$	$c_{\text{seed}} \geq$	$\sup u_0 e^{2\pi\eta}$ on disc
1/6	R_0 , right	$2 \cdot 10^{-5}$	0.006	0.0682	$3.4 \cdot 10^{-7}$	0.7628	$0.51921 \leq 0.52$
4/5 ($= \frac{1}{5}$ left)	R_0	$5 \cdot 10^{-4}$	0.0257	0.5332	$4.5 \cdot 10^{-4}$	0.4520	$0.50040 \leq 0.52$
5/6 ($= \frac{1}{6}$ left)	R_1	$5 \cdot 10^{-4}$	0.026	0.4308	$8.4 \cdot 10^{-6}$	0.4600	$0.58953 \leq 0.59$ (on $r' = 2.5 \cdot 10^{-4}$)
1/7 (not used)	–	$5 \cdot 10^{-6}$	0.00375	0.0483	$4.8 \cdot 10^{-4}$	0.5914	–

The last column is the Δ -hypothesis of P1e Lemma H(4) with the region base. For 5/6 it holds only on $r' = 2.5 \cdot 10^{-4}$, not on the full certified radius $5 \cdot 10^{-4}$. So the seed 5/6 (i.e. $\frac{1}{6}$ from the left) is used only on the disc of radius $r' = 2.5 \cdot 10^{-4}$, in Theorem M' as well as in Theorem main'. The covering disc $(\frac{1}{6} - 0.999 \cdot 2.5 \cdot 10^{-4}, \frac{1}{6})$ of item 5 is chosen accordingly.

3. Middle and exact seeds (*P1h_lemmaB.py exact*). P1e Lemma B' holds, with exact β, P_1, P_2 , exact $|Z_n|$ and env^0 , at every seed:

- R_0 : all 227 seeds with $10 \leq n \leq 150$ and radius $\varepsilon_n(y_0)$ satisfy $E \leq 0.00214$;
- R_1 : all 654 seeds with $10 \leq n \leq 300$ and radius $\varepsilon_n(y_1)$ satisfy $E \leq 0.0136$ (the maximum is at $2/13$);

$R \leq \frac{1}{2}$ throughout, and all side conditions pass.

4. Deep seeds (*P1h_lemmaB.py crude*). P1e Lemma B holds with the region data of Table 1 and the induction hypothesis $|Z_n| \geq 0.02e^{-0.005n}$:

- on R_0 for $151 \leq n \leq 2 \cdot 10^5$, with $\max E \leq 1.5 \cdot 10^{-10}$;
- on R_1 for $301 \leq n \leq 2 \cdot 10^5$, with $\max E \leq 0.438$ (at $n = 301$), $R \leq 0.125$. The script also prints $\log E$ (Arb upper bound, no underflow): $\log E(3000) \leq -203.41$, so $E(3000) \leq 4.56 \cdot 10^{-89}$ (*lemmaB_R1_crude.txt*), and $\log E(2 \cdot 10^5) \leq -15388.5$, so $E(2 \cdot 10^5) \leq e^{-15388} < 10^{-6682}$ (*lemmaB_R1_crude_20001_200000.txt*, regenerated with $P1H_Y = 0.59$, $P1H_G = 0.5762$, $P1H_ETA = 0.0037$, which the PASSED line now prints).

5. Direct covering (*P1h_caseA.py*; Proposition 3). $[x_1, \frac{1}{6}] \setminus \mathcal{D}_1$ and $[\frac{1}{6}, \frac{1}{5}] \setminus \mathcal{D}_0$ are covered with $|Z^0|, |Z^\pm| > 0.05$ and $\text{margin} > 0.05$:

- R_1 : 1023 intervals, $\min |Z| \geq 0.2959$;
- R_0 : 1939 intervals, $\min |Z| \geq 0.5215$, $\min |Z^\pm| \geq 0.2907$, $\text{margin} \geq 0.05217$, $|\omega| \leq 1.875$.

The discs are the hazards of height $10 \leq n \leq 59$ (region base), $(\frac{1}{6}, \frac{1}{6} + 0.999 \cdot 2 \cdot 10^{-5})$, $(\frac{1}{5} - 0.999 \cdot 2.5 \cdot 10^{-4}, \frac{1}{5})$ and $(\frac{1}{6} - 0.999 \cdot 2.5 \cdot 10^{-4}, \frac{1}{6})$.

Proposition 3 (A', refined covering with $M > 59$; PROVED). *Let x lie in no hazard of height ≥ 10 and in no low-seed disc, and let $N \geq 151$. Write $\Phi = \Phi_{<60} + \Phi_{\geq 60}$ by the degree of the a_m . Let ψ'_n be the coefficients of $e^{\Phi_{<60}}$, $A = \sum_{m < 60} \sup |a_m|$, and c_n the coefficients of $\exp(\sum_{m < 60} \sup |a_m| Y^m)$. Then for every $M \geq 0$ and $s \in \{0, \pm 1\}$:*

$$\left| Z_N^s - \sum_{n \leq M} \psi'_n \zeta^{-n^2 - sn} \right| \leq e^A (e^{\tau/59} - 1) + \left(e^A - \sum_{n \leq M} c_n \right).$$

Proof. By Lemma 1 (P1e H(3) with base y), $\sum_{m \geq 60} |a_m| \leq \tau/59$, so the coefficient sum of $e^{\Phi_{\geq 60}}$ is at most $e^{\tau/59}$. Since $\psi = \psi' * \chi$ with χ the coefficients of $e^{\Phi_{\geq 60}}$ and $\chi_0 = 1$, we get $\sum |\psi - \psi'| \leq (\sum |\psi'|)(\sum |\chi| - 1) \leq e^A (e^{\tau/59} - 1)$. Also $\sum_{n > M} |\psi'_n| \leq \sum_{n > M} c_n$. For $M \leq 59$ this is P1g Proposition A $^\pm$. Allowing M up to 400 matters near the seeds: the resonant coefficient a_q is large there. $\square$

Theorem 4 (M'; PROVED, computer-assisted). *For all $N \geq 1$ and $\gcd(a, N) = 1$ with $a/N \in [0.143, 0.857]$: $|Z_N(a/N)| \geq 0.02 e^{-0.005N}$.*

Proof. On $[\frac{1}{5}, \frac{4}{5}]$ this is P1e Theorem M. By conjugation it remains to treat $x \in [0.143, \frac{1}{5})$. The proof is the strong induction of P1e Theorem M, case by case, with the data of Proposition 2 in place of P1e's.

- Base $N \leq 150$: item 1 (on R_1 , $N \leq 300$).
- Deepest hazard $n \geq 151$ (R_0) or $n \geq 301$ (R_1). Use item 4 and Lemma 1(2). Then $|Z_N| \geq \frac{1}{2} \cdot \frac{1}{2} \sqrt{\pi/2A} c_0 e^{-\kappa n} e^{-n|F(t_0)|N}$ with $n|F(t_0)| \leq g^n/(5n) + n\pi R_0^2 \leq y^n \leq 0.59^{151}$ and $N \geq 1/(n|d|) \geq 0.08/(ny^n)$. This is $\geq c_0 e^{-\kappa N}$.
- Deepest hazard $10 \leq n \leq 150$ (R_0) or ≤ 300 (R_1). Use item 3: $|Z_N| \geq \frac{1}{2} \cdot 0.99 \cdot \frac{1}{2} \cdot 0.545 e^{-10^{-4}N} \geq c_0 e^{-\kappa N}$.
- Low-seed disc, no hazard. Use item 2 (for 5/6 only on $r' = 2.5 \cdot 10^{-4}$): $|Z_N| \geq 0.452 e^{-4.5 \cdot 10^{-4}N}$.
- Otherwise. Use item 5 and Proposition 3: $|Z_N| \geq 0.2959$.

The cases are exhaustive, exactly as in P1e. $\square$

3 The P1g conditions on $[\frac{1}{6}, \frac{5}{6}]$

Theorem 5 (main'; PROVED, computer-assisted). *P1g Theorem main holds on $[\frac{1}{6}, \frac{5}{6}] \setminus \{\frac{1}{2}\}$ with the same constants: $|Z_N^\pm| \geq 0.02 e^{-0.005N}$, $|\omega| \leq 7.96$, $\min_i |G_i| \geq 0.0499$.*

Proof. On R_0 run the P1g induction with the data below.

- Base: item 1.
- Twisted low seeds (P1g_lowseed.py):
 - 1/6: $r' = 2 \cdot 10^{-5}$, $\max B_\pm \leq 0.111$, $c^\pm \geq 0.4661$, $\text{margin} \geq 0.4239$, $|\omega| \leq 0.8984$;

– 4/5: $r' = 2.5 \cdot 10^{-4}$, $\max B_{\pm} \leq 0.0656$, $c^{\pm} \geq 0.5628$, $\text{margin} \geq 0.1817$, $|\omega| \leq 1.254$.

- Exact Lemma $B^{\pm}$ at the 227 seeds: $\min(\text{margin}_n - \sigma) \geq 0.1713$.
- Crude Lemma $B^{\pm}$ for $n \geq 151$: $\sum_{151 \leq n \leq 2 \cdot 10^5} \sigma(n) \leq 5.6 \cdot 10^{-8}$ and $\sum \rho(n) \leq 3.8 \cdot 10^{-9}$. The tail $n > 2 \cdot 10^5$ is handled as in P1g Lemma $B^{\pm}$ (iii) with $\theta = 0.7544$.
- Covering: item 5.

A chain of deepest hazards starting in R_0 stays in R_0 (Lemma 1). So the margin is $\geq 0.05 - 5.6 \cdot 10^{-8} - 10^{-50} \geq 0.0499$, and $|\omega| \leq \max(1.875, 1.254, 1.039)(1 + 10^{-8})$. $\square$

4 Theorems B' , C' : the landscape at general ℓ

Lemma 6 (LS(ℓ_1); PROVED for $(\ell_1, N) = (\log 2, N \geq 16)$ and $(0.5522, N \geq 17)$, computer-assisted). *P1f Lemmas LS, LSp and head hold for every ζ with $\ell \in [\ell_1, \log 4]$ and $N \geq N_1$. The margins are as printed by `P1h_landscape.py`; at $\ell_1 = \log 2$, $N = 16$: $T - W_{D<0} \geq 0.050$, $D \geq 0.070$, head margin ≥ 0.076 , $\frac{\pi}{4} - |\theta^*| \geq 0.27$.*

Proof. Each estimate in the proofs of P1f Lemmas LS, LSp, head is an explicit expression in (ℓ, N, θ^*) . The quantities that are compared with the tie height are:

- $T - 2\delta\ell$, $T - W_{D<0}$, $T - H_{\text{head}}$, $D = T - \frac{\pi^2}{6N^2} - 2\delta\ell$, and $D' = \frac{\ell^2}{4} - 2\delta\ell$.
- Here $T = \frac{\ell^2}{4 \cos \theta^*} - \frac{\pi^2 s^2 \cos \theta^*}{4N^2} + \text{Re}(E_w e^{-i\theta^*})$ and $\delta = \frac{1}{8N}$.
- Each has the form $\frac{\ell^2}{4} - \frac{\ell}{4N} + (\ell\text{-free})$, which increases in ℓ for $\ell > \frac{1}{2N}$.
- The Taylor factor $1/(e^{2N \text{Re } y} - 1)$ with $\text{Re } y \geq \ell/4$ decreases in ℓ .

In detail, each such margin is $Q(\ell) = \frac{\ell^2}{4} - \frac{\ell}{4N} + C$ with C independent of ℓ , and for $\ell \geq \ell_1$

$$Q(\ell) - Q(\ell_1) = (\ell - \ell_1) \left(\frac{\ell + \ell_1}{4} - \frac{1}{4N} \right) \geq (\ell - \ell_1) \left(\frac{\ell_1}{2} - \frac{1}{4N} \right) \geq 0, \quad \text{since } \ell_1 \geq 0.55 > \frac{1}{2N} \ (N \geq 16).$$

Likewise $\text{Re } y \geq \ell/4 \geq \ell_1/4$ gives $1/(e^{2N \text{Re } y} - 1) \leq 1/(e^{N\ell_1/2} - 1)$. So these are evaluated at $\ell = \ell_1$. The size bounds ($|L_\mu|$, Δ , $|x - x_\mu| \leq 15$) increase with ℓ and are evaluated at $\log 4$. The angle is $|\tan \theta^*| \leq \pi s/(N\ell_1)$. The two places where P1f used a frozen constant are replaced by the exact quantity: the ray coefficient 0.34 by $\cos(|\theta^*|_{\max} + \frac{\pi}{4})$, and $\eta_N = \min(16.2/N^2, 0.1)$ by $\min((2\pi^2 s^2 \cos \theta^* - C_R)/N^2, T - W_{D<0})$. `P1h_landscape.py` evaluates all of them in Arb for $N_1 \leq N \leq 10^5$ (all positive). For $N > 10^5$ the argument is by hand. Each height margin is its limit $\frac{\ell^2}{4} \geq \frac{\ell_1^2}{4}$ minus the N -dependent corrections $\frac{\ell}{4N}$, $\frac{\pi^2}{6N^2}$ and the head term $\frac{2\delta}{N} \log \frac{1}{1.8\delta}$ ($\delta = \frac{1}{8N}$). For $N > 10^5$ and $\ell \leq \log 4$:

$$\frac{\ell}{4N} + \frac{\pi^2}{6N^2} + \frac{\log(8N/1.8)}{4N^2} \leq 3.5 \cdot 10^{-6} + 1.7 \cdot 10^{-10} + 3.4 \cdot 10^{-10} < \frac{0.55^2}{8} \leq \frac{\ell_1^2}{8},$$

so every height margin is $\geq \ell_1^2/8 > 0$. For the N^{-2} margins, $|\tan \theta^*| \leq \pi s/(N\ell_1) \leq \pi s/(0.55 \cdot 10^5)$ gives $\cos \theta^* \geq 0.99$ (for the s used), and

$$\frac{2\pi^2 s^2 \cos \theta^* - C_R}{N^2} \geq \frac{2\pi^2 \cdot 0.99 - 1.65}{N^2} > \frac{17.8}{N^2} > 0 \quad (s \geq 1).$$

The head lemma's constant 0.0837 requires $N \geq 16$, which is why $N_1 \geq 16$. $\square$

Theorem 7 (B', C'; PROVED, computer-assisted through Theorems 4, 5). 1. *P1f Theorem B (zeros of B accumulating at ζ along the tie ray; simple; simple poles of G_0^T) holds for every $\zeta = e(a/N)$ with $a/N \in [0.143, 0.857]$ and $N \geq 17$, and with $a/N \in [\frac{1}{6}, \frac{5}{6}]$ and $N \geq 16$.*

2. *P1f Theorem C(i)–(iii) holds for $a/N \in [\frac{1}{6}, \frac{5}{6}]$, $N \geq 16$.*

*Consequently the poles of G_0^T accumulate on the whole arc $\arg q \in [0.286\pi, 1.714\pi]$. The poles of U and V accumulate on $\arg x \in [\frac{\pi}{6}, \frac{5\pi}{6}] \cup [\frac{7\pi}{6}, \frac{11\pi}{6}]$ (standing *thm:model*, *thm:RJ*).*

Proof. The proof of P1f Theorems asym, B, C uses $\ell \geq \ell_0$ only through Lemmas LS, LSp, head (now Lemma 6), through $A(n) \neq 0$, i.e. $Z_N \neq 0$ (Theorem 4), and, for C, through $Z_N^+ \neq 0$ and the ω -conditions (Theorem 5). The EM lemma, the class decomposition and Rouché are ℓ -free, given $|u_0| < 1$ and $N\ell \geq 3$ (Lemma W). $\square$

Table 2: $N_1(\ell_1)$: least N with all landscape margins positive (`P1h_landscape.py scan`). The $\ell_1 = \log 2$ and 0.5522 rows are PROVED ($N \leq 10^5$ in Arb plus the asymptotic remark). The others are checked on the window $[N_1, N_1 + 4000]$ only (VERIFIED).

ℓ_1	log 2	0.6	0.5	0.4	0.3	0.2	0.12	0.1	0.05	0.02
N_1	13 (16)	15 (16)	18	24	34	55	105	132	311	947
$N_1\ell_1$	9.0	9.0	9.0	9.6	10.2	11.0	12.6	13.2	15.6	18.9

So the tie-ray analysis needs only $N\ell \gtrsim 10$. The slow growth comes from the head term $\frac{2\delta}{N} \log \frac{1}{1.8\delta} \sim \frac{\log N}{4N^2}$. **Conditional statement (PROVED modulo the window check for $\ell_1 \notin \{\log 2, 0.5522\}$):** for every ζ with $\ell \geq \ell_1 > 0$, $N \geq \max(16, N_1(\ell_1))$ and $Z_N(a/N) \neq 0$, P1f Theorem B holds at ζ . The only missing input on the full arc $\ell > 0$ is $Z_N \neq 0$.

5 The obstruction as ℓ decreases

The P1e induction has three components: seeds (low and exact), hazards (Lemma B crude), and the direct covering. Each degrades in a quantifiable way.

1. Golden threshold (PROVED as a statement about Lemma B). The crude Lemma B controls the error at a deep seed j/n by $\sqrt{2n} \sup_k |e^{B_k}| \leq \sqrt{2n}(1-y)^{-n/2}$ (triangle inequality over k) against $|Z_n| \geq c_0 e^{-\kappa n}$ and the hazard size $d \leq 12.5y^n$. So $E, R \asymp n^{5/2}\theta^n$ with $\theta = e^\kappa y / \sqrt{1-y}$, and one needs $g < y$ and $\theta < 1$. That is

$$y < y^* = 0.61633 \quad (\kappa = 0.005; \frac{\sqrt{5}-1}{2} \text{ as } \kappa \rightarrow 0), \quad \text{i.e. } \ell > 0.48398, \quad x > 0.132948.$$

For $x \leq 0.13295$ no choice of (y, η) makes the crude Lemma B hold at any height. This is the sharp form of P1e's " $\ell > 0.48$ " remark.

2. Crossover height (VERIFIED, `P1e_lemmaB.py`). Below n_0 the exact data (Lemma B') must be used:

g (region left end)	0.5 ($\frac{1}{6}$)	0.5762 ($\frac{1}{7}$)				
y	0.52	0.54	0.56	0.59	0.60	0.61
n_0	67	89	126	301	514	1467

$n_0 \sim \log(C)/\log(1/\theta)$ with $\theta \rightarrow 1$ as $y \rightarrow y^*$. The exact seeds up to n_0 number $\approx 0.3 n_0^2 |\text{region}|$, at cost $O(n^2)$ each. The region $[0.1335, \frac{1}{7}]$ ($g \in [0.576, 0.615]$) would need n_0 in the thousands, and near $x = 0.133$ it needs $n_0 \rightarrow \infty$. HEURISTIC estimate: about 10^4 – 10^6 Arb seed evaluations for $[0.135, \frac{1}{7}]$, a Rust job, not attempted.

3. Low-seed/covering handover (VERIFIED). Two constraints conflict near a low seed p/q :

- The seed certificate needs $\sup |u_0|e^{2\pi\eta} \leq y$, so $\eta \leq \log(y/g)/2\pi$. At $\frac{1}{7}$ with $y = 0.59$ this gives $\eta \leq 0.00376$.
- With that η , P1e Lemma O certifies only $r = 5 \cdot 10^{-6}$. At 10^{-5} it gives $B = 0.989$, and larger radii fail the gap condition $d \leq 2 \text{ gap}$.

The direct covering (Proposition 3) needs $A = \sum \sup |a_m|$ moderate, but at distance d from p/q , $|a_q| \approx g^q/(2\pi q^2 d)$. At $d = 5 \cdot 10^{-6}$ from $\frac{1}{7}$ this is 13.6, and the bound is $e^A(e^{\tau/59} - 1) = 754$ even with $M = 800$ (`caseA.R1.1.7.txt`, FAILED as expected). So the disc and the covering do not meet at $1/7$. The arc stops at $x_1 = 0.143 = \frac{1}{7} + 1.43 \cdot 10^{-4}$, where $|a_7| \approx 0.48$ and the covering passes. A larger y near $\frac{1}{7}$ (e.g. 0.60, $\eta \leq 0.0064$) would enlarge the disc, at the price of $n_0 = 514$ (item 2).

4. The ω -margin (P1g) fails first (computer-assisted diagnosis). On R_1 , Theorem M' closes, but the P1g ω -induction does not, for two reasons:

- At the seed $2/13$ the exact twisted Lemma B $^\pm$ gives $E_\pm \leq 0.332$, hence $\rho \leq 3.60$ and a loss 3.39 against a margin 0.58. This is the only one of the 654 seeds that fails.
- The crude chain has $E_\pm \leq 0.438$ at $n = 301$, $\rho(301) \leq 9.75$ and $\sum \sigma \leq 1050$. The ω -induction needs $\sum \sigma < 0.05$, which requires $E \lesssim 10^{-3}$ at the first crude height, i.e. exact seeds to $n \approx 500$ at $y_1 = 0.59$.

Hence U, V density is proved only on $[\frac{1}{6}, \frac{5}{6}]$, not on $[0.143, 0.857]$ (OPEN there). The covering itself passes on all of $[x_1, \frac{1}{6}]$ with margin ≥ 0.0507 (VERIFIED as part of the Z-certificate runs).

5. Below $\ell \approx 0.484$: the moment hypothesis (HEURISTIC/OPEN). The failing input is the single inequality $\sup_k |e^{B_k(t)}| \leq (1-y)^{-n/2}$, used with $\sum_k |\gamma_k|$, i.e. the loss factor $\Lambda = \text{env}/|Z_n|$ of P1e §2.

- On drift families Λ grows like $e^{0.008q}$ (P1e: $\Lambda \leq 2.7 \cdot 10^7$ at $182/2001$).
- The moment ratios stay bounded: $D_1 \leq 0.05$, $D_2 \leq 0.003$.
- The replacement hypothesis would be $|f^{(j)}(t)| \leq C_j(2\pi n)^j |f(0)|$ on $|t| \lesssim 1/n$ for $f(t) = \sum_k \gamma_k e^{\Phi_n(\zeta_0^k e(-t))}$. Carried through Lemma lap, it would give an error $\approx D_2(2\pi n)^2 d$ with no Λ , and hence a threshold $y < 1$, i.e. all of $\ell > 0$.

The hypothesis is not inductively closed by anything proved here. The derivative f' at level N is not controlled by level- n moments without a second-order version of Lemma R, which is not available (OPEN). The exact seed data remain healthy well below the threshold: on $[\frac{1}{7}, \frac{1}{6})$, $|Z_n| \geq 0.545$ for all $n \leq 300$, with a slow drift $0.593 \rightarrow 0.545$ from $n \leq 150$ to $n \leq 300$ (VERIFIED). So the obstruction is in the method, not in the object.

6 What is not proved; weakest points

1. **Review scope.** Reviewer AD judged the note correct with minor gaps (addressed here). Not re-derived independently: Lemma 1 (region-dependent bases), Proposition 3 ($M > 59$), and the transcription of the P1f landscape estimates into `P1h.landscape.py` were checked by one reviewer, not re-derived line by line. The landscape script re-derives the P1f bounds as functions of (ℓ, N) ; an error there would affect Theorem B' only, not Theorem M'.

2. **Hand tails.** The crude Lemma B is machine-checked to $n = 2 \cdot 10^5$. Beyond that $E, R \leq Cn^{5/2}\theta^n$ with $\theta \leq 0.926$ by P1e's term-by-term argument, not machine-checked. The margin is absurd: $E(2 \cdot 10^5) \leq e^{-15388} < 10^{-6682}$ (printed log-bound).
3. **Tight Δ -hypothesis at $5/6$.** $\sup |u_0|e^{2\pi\eta} = 0.58953 \leq 0.59$ on the disc of radius $2.5 \cdot 10^{-4}$. It is rigorous, but it has only $5 \cdot 10^{-4}$ of room. (On the full $5 \cdot 10^{-4}$ disc it would exceed 0.59, so only $r' = 2.5 \cdot 10^{-4}$ is used.)
4. **Monotonicity in g .** The crude region bound uses one g per region, as P1e does, and relies on P1e's statement that all terms increase with $|d|$ and g . This is inherited, not re-proved.
5. **Inherited standing.** P1d (Lemmas W, 6, 8; Theorems 2, 5, 9), P1e (H, R, O), P1f ($O(t)$ and j_0 not explicit), P1g, python-flint/Arb. For U, V also `thm:model`, `thm:RJ`.
6. **Not reached.** $Z_N \neq 0$ on $(0.0804, 0.143)$: OPEN. The target $[\frac{1}{6} + \delta, \frac{5}{6} - \delta]$ of the brief is exceeded (on ℓ : from 0.855 to 0.552), but the gap to the golden threshold 0.484 and to $\ell = 0$ remains.

Files (rootunity_zeros/general/P1h/):

- Scripts: `P1h_seeds.py`, `P1h_lemmaB.py` (environment variables `P1H_Y`, `P1H_G`, `P1H_ETA`, `P1H_NOMARGIN`), `P1h_caseA.py` (`P1H_Y`, `P1H_ZONLY`), `P1h_lowseed.py`, `P1h_hazard_check.py`, `P1h_landscape.py`.
- Outputs: `seeds_R0*.txt`, `seeds_R1*.txt`, `lemmaB_R*.txt`, `caseA_R*.txt`, `caseA_discs_R*.txt`, `low*.txt`, `landscape*.txt`.
- Summary: `P1h_certlog.txt`.

Every run used `perl -e 'alarm 290; exec @ARGV'` and stayed under 300 MB. A fresh check gave 24 GB RAM and 7.9 GB free disk; P1h writes under 0.5 MB. No Rust was needed at this range. Rust is the natural tool for item 2 of §5.